

Multi-Stage Water Rocket

Event Description:

The Multi-Stage Water Rocket Competition is designed to provide participants with an opportunity to showcase their skills, creativity, and teamwork. the event is a fun and practical event where participants design and launch rockets made of plastic bottles using water and air pressure. It helps participants understand the basics of physics, aerodynamics, and propulsion. while adhering to fair play and safety standards.

Eligibility Criteria:

- Open to all students from any department and college.
- Participation is team-based.
- Prior registration is mandatory
- Basic knowledge/skills related to the event is recommended but not mandatory.

Rules & Regulations:

1. Participation Guidelines

- Team size: 2-4 members per team
- Only approved tools, instruments, or resources may be used.
- Prohibited items or practices are not allowed.
- All participants must follow safety and discipline protocols.
- Models must conform to stated size, safety, and material restrictions.
- All teams present a brief (max 5 min) oral explanation.
- Participants must follow laboratory safety and hygiene protocols.

2. Match/Performance Rules:

- Design Rules:
 - Maximum bottle size: standard 2 liters.
 - No use of harmful materials, chemicals, or metallic parts.
 - Fins and nose cones may be added using safe, lightweight materials.
- Launch Procedure:
 - Rockets must be launched using the provided launcher.
- The event will be conducted in performance-based format
- Each round continues until a clear winner/qualifier is determined.
- Elimination occurs if: Participants fail to meet the basic event requirements.
- They violate time limits or performance rules.
- They are deemed unfit to continue by judges/referees.

3. Evaluation/Scoring Criteria:

- Maximum height achieved within three given attempts .
- Time of flight
- Accuracy of landing
- Creativity and stability of Design.

4. Disqualification Factors

- Exceeding defined limits (time, size, capacity, etc.).
- Use of prohibited items or unfair practices.
- Intentional violation of event rules.
- Unsportsmanlike behavior or misconduct.